DESIGN AND IMPLEMENTATION OF INTERLEAVED BOOST CONVERTER WITH VOLTAGE MULTIPLIER CIRCUIT FOR SOLAR – GRID INTEGRATION

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Abstract: An interleaved boost converter with voltage multiplier circuit for solar and grid integration is shown in this article. The solar panel's electricity may now be integrated into the grid thanks to this converter. Two stages of this converter: the "Interleaved Boost Stage" and the "Voltage Multiplier Circuit". Input DC voltage is changed to boosted AC voltage in the interleaved boost stage. High voltage DC is produced by converting AC current via a specialized rectifier known as a voltage multiplier circuit. A prototype is then created to verify the underlying performances after the converter is first simulated using the MATLAB/Simulink platform.

Keywords: — Boost converter, Interleaved Boost Stage, Voltage Multiplier Circuit, Solar – Grid Integration.

1 Introduction

The power electronics interface is important in many applications and topologies. It makes it possible to convert power from renewable energy sources effectively. A photovoltaic (PV) system's output varies according on the amount of sun irradiance, making it an unreliable source of renewable energy. The PV system needs a power electronic interface to maintain its reliability. [1-2] discusses the benefits of different power electronic interface topologies. Conventional boost converters have various shortcomings. Use of it in high-power applications is not advised. because in order to obtain high voltage gain, it needs to run at a higher duty ratio. Also, it draws a lot of current from the input source at extreme duty cycles, which could damage the converter's components. Interleaving topology is used to get around these issues. In [3-4], the conventional step-up topology and interleaved topology are compared and contrasted. By embracing a cascading design, boost converters may be coupled in series to provide noteworthy voltage gain at minimal duty cycles. A comparative study of the cascaded boost converter is shown in [5–6]. However, it has the same drawbacks as the standard boost converter. By connecting two converters in parallel, the input ripple, stress on the switches, and other issues can be reduced. In [7], this kind of topology is talked about. The output gain, however, is the same

as that of a typical boost converter, which is a problem. High ratios of conversion can also be attained by using a switched inductor [8–9]. Because there is leakage inductance, voltage spikes are the major problem. Therefore, a second clamping circuit is needed to correct the voltage spikes. For a high conversion ratio, some converters use switched capacitors. High power density, a reduction in pressure across switches, and other advantages are available [10–11]. However, there are drawbacks, including the requirement for more active switches and losses brought on by voltage variations in the capacitors. It is also difficult to develop a gate driver that can turn these active switches ON or OFF. Avoid using convoluted topologies if possible. Utilizing isolated topologies such as flyback, push pull, forwards, full bridge, and others, high conversion rates may be achieved [12–15]. An isolated topology's input and output components are separated using a transformer. However, it also experiences leaking inductance, which lessens the power density of the converter overall. A voltage multiplier cell integrated with typical boost converter provide high voltage gains [16]. There might, however, be substantial input current ripples.

In this study, a voltage multiplier cell-equipped non-isolated interleaved boost converter for solar-grid integration is developed. The voltage obtained from the solar panel is increased using this converter. The suggested work's block diagram is shown in Figure 1. The interleaving architecture offers multiple benefits over traditional converters, including minimized input ripple, reduced voltage stress across active switches, high voltage gain, and high efficiency. These benefits promote more wise choices for circuit components. Using a voltage multiplier circuit enhances the voltage gain.

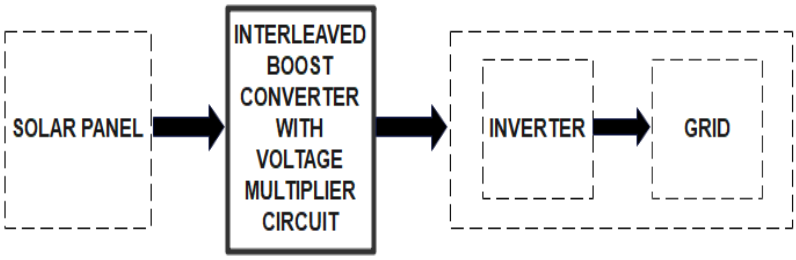

Fig 1: Block diagram of the proposed work

The rest of the document is organized as follows: In part II, the mechanism of action is covered. The gain equation, the volage equations, and the component designing are all determined in this section after each mode has been reviewed. The converter's waveforms and their simulated part are discussed in section III. The implementation of the hardware prototype is shown in Section IV. Hardware setup, PCB layout, and gate pulse generation are all clarified in this section. At last, the end is introduced in Segment V.

2 MODE OF OPERATION

The voltage multiplier cell and interleaved boost stage are the two stages of this converter. The converter's circuit design is seen in Figure 2. Two inductors are switched by two active switches in the interleaved boost stage. Figure 3 illustrates how two gate signals with a 180o phase shift between them control the active switches. Diodes and capacitors make up the voltage multiplier circuit, which is utilized to correct the interleaved boost stage's AC voltage.

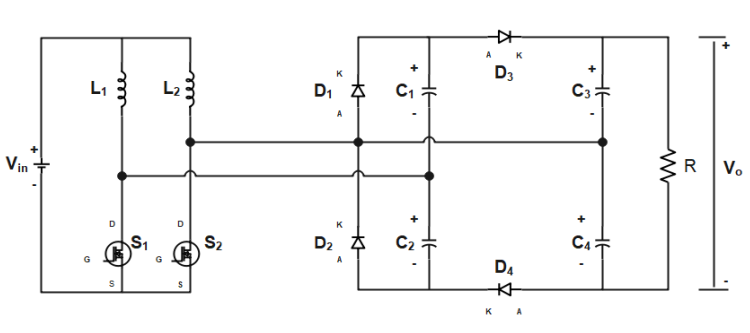

Fig 2: Circuit Diagram of the Converter

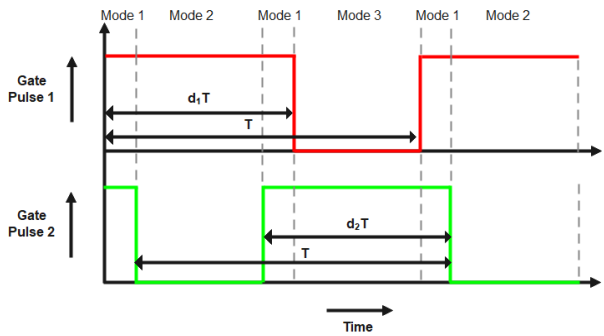

Fig 3: Gate Pulse Sequence

A. Mode 1:

Both switch 1 and 2 are under ON stage: In this state, each and all diodes are reverse biased and off, while both inductors are being charged by the source. Figure 4 depicts the analogous circuit for this mode.

$$V_{in} = L_1 \frac{di_{L1}}{dt} \tag{1}$$

$$V_{in} = L_2 \frac{di_{L2}}{dt} \tag{2}$$

$$V_O = V_{C3} + V_{C4} \tag{3}$$

B. Mode 2:

Switch 1 is ON and Switch 2 is OFF: L1 is charging from the supply, L2 is delivering stored energy to the voltage multiplier circuit. The forwards biased diodes D2 and D3 are. Figure 5 depicts the analogous circuit for this mode.

$$V_{in} = L_1 \frac{di_{L1}}{dt} \tag{4}$$

$$V_{L2} = V_{C2} - V_{in} \tag{5}$$

$$V_{in} + V_{L2} + V_{C1} - V_{C3} = 0 \tag{6}$$

$$V_{in} + V_{L2} + V_{C1} - V_O + V_{C4} = 0 \tag{7}$$

C. Mode 3:

Switch 1 is OFF and Switch 2 is ON In this mode, the inductor L1 transfers its stored energy to the voltage multiplier circuit while the inductor L2 continues to charge from the supply. The forwards biased diodes D1 and D4 are. Figure 6 depicts the analogous circuit for this mode.

$$V_{in} = L_2 \frac{di_{L2}}{dt} \tag{8}$$

$$V_{L1} = V_{C1} - V_{in} \tag{9}$$

$$V_{in} + V_{L1} - V_{C4} + V_{C2} = 0 \tag{10}$$

$$V_{in} + V_{L1} + V_{C3} - V_O + V_{C2} = 0 \tag{11}$$

Gain Equation:

Applying the energy balance concepts to the voltage equations derived from the three modes yields the gain equation for the converter.

Applying energy balance concept for inductor L1,

$$\int_0^{T_{ON}} V_{L1}dt = \int_0^{T_{OFF}} V_{L1}dt$$

By substituting (1) and (9),

$$\int_0^{DT} V_{in}dt = \int_0^{(1-D)T} \left(V_{C1} - V_{in}\right)dt$$

$$V_{in}DT = \left(V_{C1} - V_{in}\right)\left(1-D\right)T$$

$$V_{in}D = V_{C1}\left(1-D\right) + V_{in}D - V_{in}$$

$$V_{C1} = \frac{V_{in}}{1-D} \tag{12}$$

Similarly for inductor L_2,

$$\int_0^{T_{ON}} V_{L2}dt = \int_0^{T_{OFF}} V_{L2}dt$$

By substituting (2) and (5),

$$\int_0^{DT} V_{in}dt = \int_0^{(1-D)T} \left(V_{C2} - V_{in}\right)dt$$

$$V_{in}DT = \left(V_{C2} - V_{in}\right)\left(1-D\right)T$$

$$V_{in}D = V_{C2}\left(1-D\right) + V_{in}D - V_{in}$$

$$V_{C2} = \frac{V_{in}}{1-D} \tag{13}$$

$$V_{C1} = V_{C2} = \frac{V_{in}}{1-D} \tag{14}$$

From eq (6),

$$V_{in} + V_{L2} + V_{C1} - V_{C3} = 0$$

$$V_{in} + V_{L2} + \frac{V_{in}}{1-D} - V_{C3} = 0$$

$$V_{L2} = V_{C3} - V_{in} - \frac{V_{in}}{1-D} \tag{15}$$

Again, applying energy balance concept for inductor L_2,

$$\int_0^{T_{ON}} V_{L2} dt = \int_0^{T_{OFF}} V_{L2} dt$$

By substituting (2) and (15),

$$\int_0^{DT} V_{in} dt = \int_0^{(1-D)T} \left(V_{C3} - V_{in} - \frac{V_{in}}{1-D} \right) dt$$

$$V_{in} DT = \left(V_{C3} - V_{in} - \frac{V_{in}}{1-D} \right) (1-D) T$$

$$V_{in} D = V_{C3} (1-D) - V_{in} (1-D) - V_{in}$$

$$V_{in} D = V_{C3} (1-D) - V_{in} + D V_{in} - V_{in}$$

$$0 = V_{C3} (1-D)_{in} - 2V_{in}$$

$$V_{C3} = \frac{2V_{in}}{1-D} \tag{16}$$

Similarly for inductor L_1,

$$V_{C4} = \frac{2V_{in}}{1-D} \tag{17}$$

From eq (3)

$$V_O = V_{C3} + V_{C4}$$

By substituting the eq (16) and (17),

$$V_O = \frac{2V_{in}}{1-D} + \frac{2V_{in}}{1-D}$$

$$V_O = \frac{4V_{in}}{1-D} \tag{18}$$

The gain equation is obtained as,

$$\frac{V_O}{V_{in}} = \frac{4}{1-D} \tag{19}$$

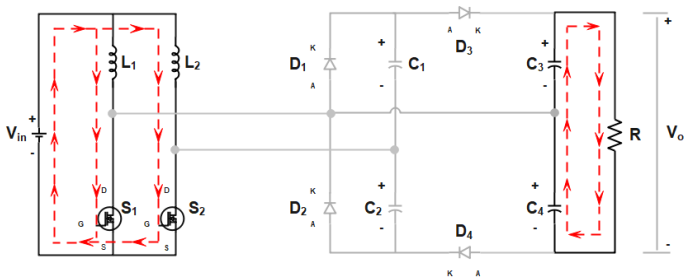

Fig 4: Mode 1 equivalent circuit

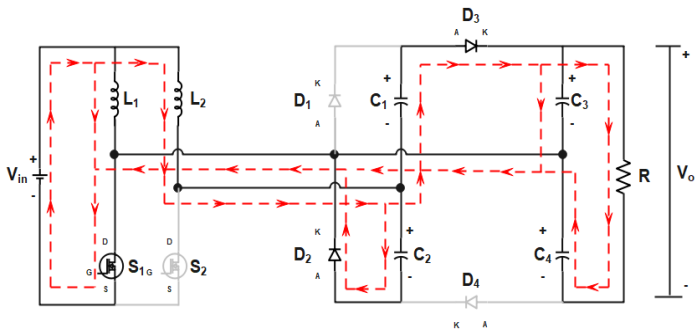

Fig 5: Mode 2 equivalent circuit

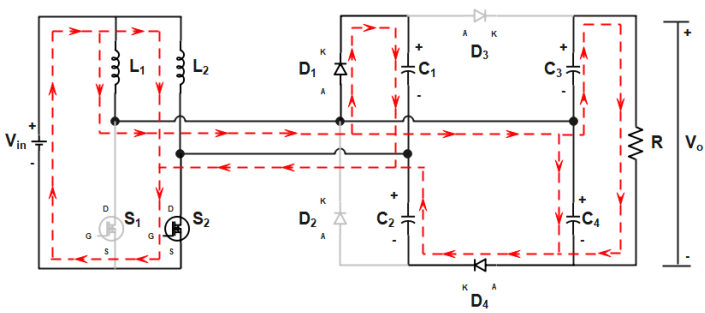

Fig 6: Mode 3 equivalent circuit

Component Design:

A. Inductor Design

For the converter to operate in continuous conduction mode, the smallest possible inductance value is given by

$$L_{1,crit} = L_{2,crit} = \frac{V_{in}D(1-D)}{I_O F_S} \tag{20}$$

Despite this, the inductor current ripples were utilized to design the inductors.

$$V_{in} = L_1 \frac{di_{L1}}{dt} = L_1 \frac{di_{L1}}{T_{ON}}$$

$$T_{ON} = L_1 \frac{di_{L1}}{V_{in}} \tag{21}$$

W.K.T

$$T_{ON} = DT$$

$$DT = L_1 \frac{di_{L1}}{V_{in}}$$

$$DT = L_1 \frac{\Delta i_{L1}}{V_{in}}$$

$$L_1 = \frac{V_{in}DT}{\Delta i_{L1}}$$

$$L_1 = L_2 = \frac{V_{in}D}{\Delta i_L F} \tag{22}$$

In this equation, the input voltage is indicated as V_{in}, D is the duty ratio, Δi_L is represented as the inductor ripple, and F is the frequency

B. Capacitor Design

The capacitor's voltage stress is established by,

$$V_{C1} = V_{C2} = \frac{V_{in}}{1-D} \tag{23}$$

$$V_{C3} = V_{C4} = \frac{2V_{in}}{1-D} \tag{24}$$

The voltage ripple and switching frequency serves as the basis for the design of the capacitor. If switching frequency is increased, ripple voltage and the size of the energy storage elements decreases. Therefore, the capacitance is determined by,

$$C = \frac{I_C}{\Delta V_C F} \tag{25}$$

Where I_C denotes the capacitor current, ΔV_C denotes voltage ripple and F denotes the switching frequency.

3 SIMULATION

Through MATLAB/Simulink, the converter simulation is carried out. Figure 7 illustrates the simulation of the converter. The input parameters for the simulation are listed in Table 1. The variation of gain with respect to duty cycle is shown in Figure 8. The output voltage is 150 VDC, using a 15 VDC input voltage. 60% of the time is in duty. Figures 9 shows the inductor current, voltage, switch voltage, diode voltage and gate pulses. L1 and L2 start charging when both switches are in the ON position and when both the switches are turned OFF, they start discharging. It is estimated that 1.13 A and 2.72 A are the maximum inductor currents respectively. When the switch is switched ON, there will be no voltage across it, creating a short circuit. When it is off, it acts as an open circuit. The maximum stress across the switch is at 38 V. Diodes 2 and 3 exhibit reverse bias during the negative half cycle, diodes 1 and 4 exhibit reverse biased during the positive half cycle. It is found that the diodes are under around 74 V of voltage stress.

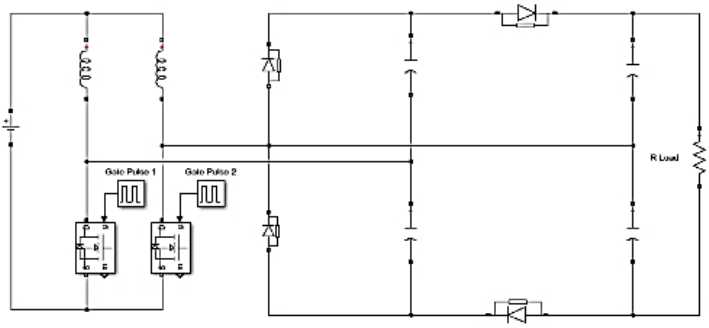

Fig 7: Simulation diagram

TABLE 1
Parameters used in Simulation

Input Voltage	15 V
Output Voltage	150 V
Switching Frequency	20 kHz
Duty Cycle	60 %
Inductors	3.75 mH
Capacitors	10 uF

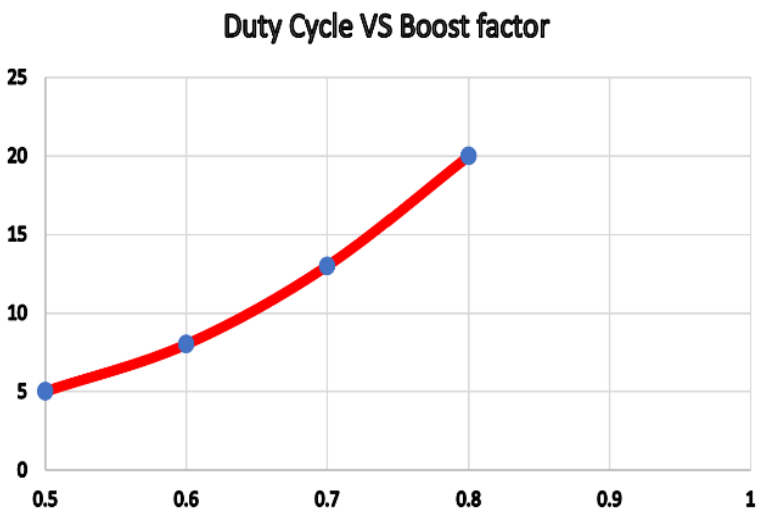

Fig 8: Variation of Gain with respect to duty cycle

When duty cycle increases correspondingly the gain of the converter increases based on the equation (19).

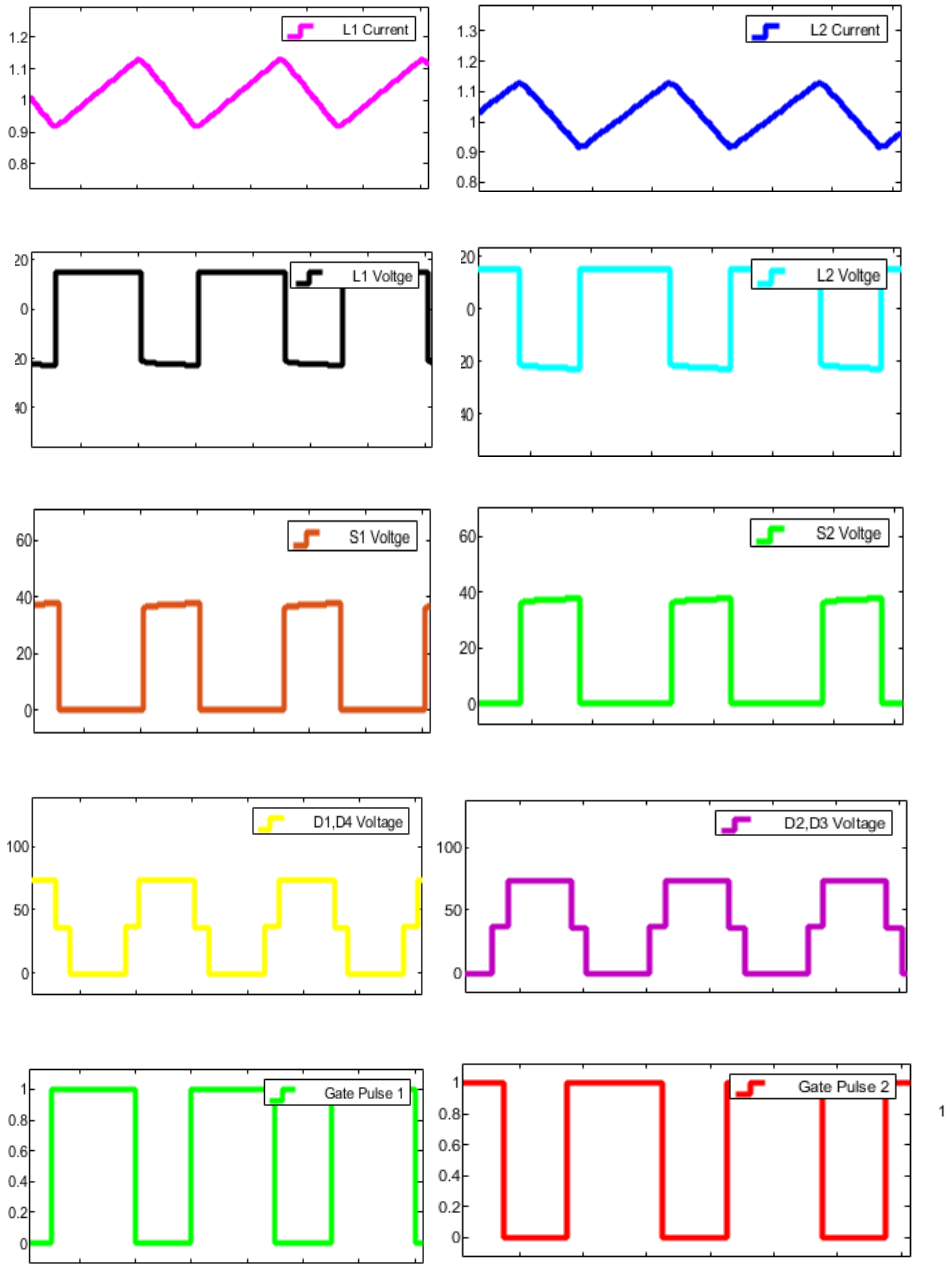

Fig 9: waveform of inductor currents, inductor voltages, switch voltages, diode voltages and gate pulses

4 HARDWARE IMPLEMENTATION

A. Gate Pulse Generation

The gate pulses for the switches are generated by using TL494 PWM IC. Figure 10 depicts the TL494 gate pulse generating circuit. An integrated circuit having a pulse width modulation controller is called the TL494. It contains sixteen pins. Power electronics applications including inverters, switch-mode power supply, and DC - DC power converters typically use it. It allows for precise beat width adjustment waveform control and takes customizable recurrence, dead time, and obligation cycle into account. It features two error amplifiers that may be used to compare a feedback signal from the output to a reference signal in to keep the output's voltage or current constant. In addition, a soft-start circuit that lowers startup inrush current is present. The TL494 can function from a broad variety of input voltages and includes an internal oscillator that may be used to create the PWM signal. It can drive power MOSFETs or bipolar transistors to control the output voltage or current. TL494 is first simulated in MULTISIM prior to creating the hardware model.

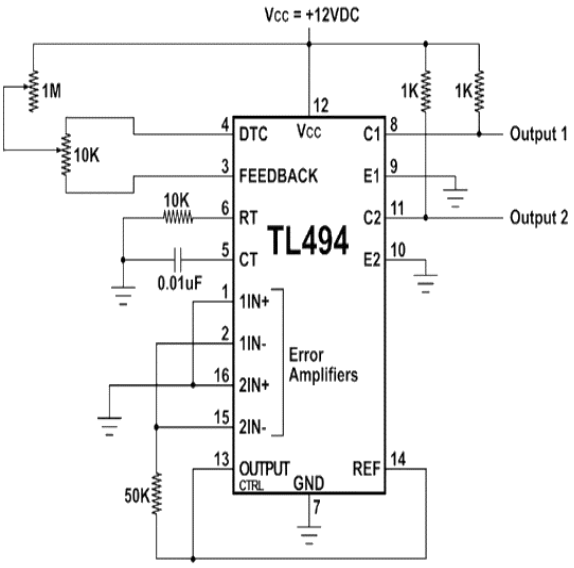

Fig 10: TL494 Circuit for gate pulse generation

TLP250 makes it feasible to electrically isolate the device's input and output. Because there is a chance that any issue on the load side would harm the gate driver. It is a low-power phototransistor optocoupler, which implies that one of its components is a phototransistor that detects the light from an infrared LED and generates an output current proportionate to the input current. A high speed photocoupler with an open collector output, the TLP250 may be utilized to drive power MOSFETs and IGBTs in a variety of applications, including motor control, power supply, and home appliances. On the TLP250 IC, a photodetector and an infrared light emitting diode are combined in a single package. The photodetector and an open gatherer yield semiconductor are located on the result side of the IC, while the driven and an info current limiting resistor are located on the info side. When a sign is put on a disc, it emits infrared light that the photodetector on the result side can detect. Figure 11 depicts the hardware model of the TL494 and TLP250, and Figure 12 depicts the outcome.

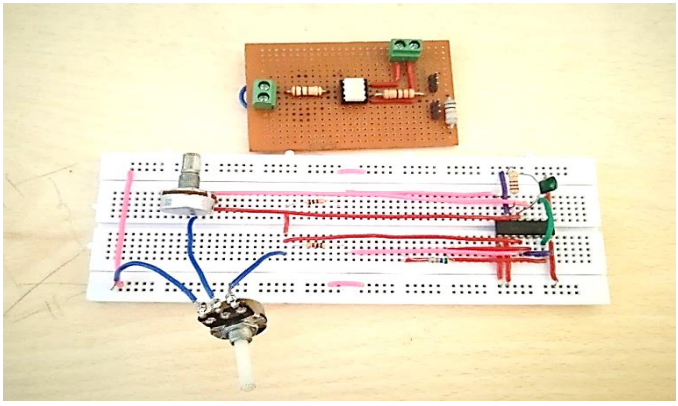

Fig 11: Hardware model of TL494 and TLP250

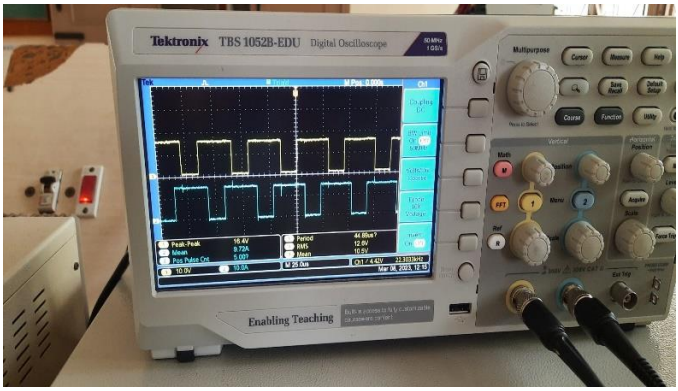

Fig 12: Results of the hardware prototype model

B. PCB Design

The primary objective of a PCB design is to develop a circuit layout that is simple to fabricate and assemble into a functional device. As a result, testing, debugging, and wiring problems are less expensive. A well-designed PCB may significantly improve the overall performance, reliability, and cost-effectiveness of electronics devices, making it a crucial step in the product development process. Figure 13 depicts the converter's PCB design.

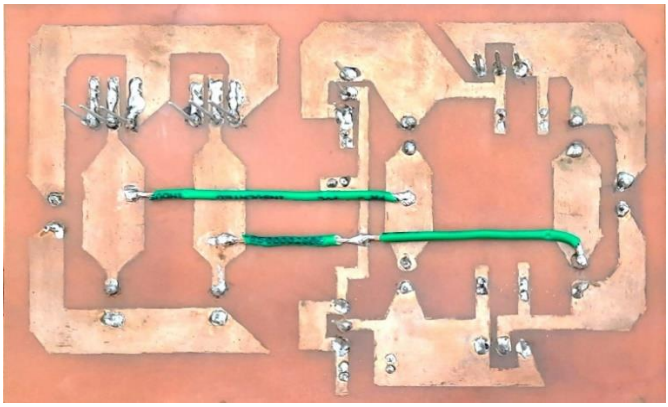

Fig 13: PCB Design

C. Hardware Setup and Experiment Results

The theory and simulation findings are used to verify a hardware prototype. Table II includes a list of the hardware prototype's components.

TABLE II
Components used in the prototype

Component	Designation	Rating
Inductors	L_1, L_2	3.75 mH, 25 A
IGBT	S_1, S_2	600 V, 80 A
Capacitors	C_1, C_2	10 uF, 250 V
Capacitors	C_3, C_4	10 uF, 450 V
Diodes	D_1, D_2	600 V, 30 A

The output voltage is 150 VDC using a 15 VDC input voltage. The switching frequency is 20 kHz, and the duty cycle is 60%. Figure 14 depicts the hardware configuration. The waveforms of gate pulses, switch voltages, inductor voltages, and inductor currents are shown in Figures 15, 16, 17, and 18. The maximum voltage stress across the switches is found to be 43.2 V. It was found that the peak inductor voltage was about 46 V. The inductor's average current, which peaks at 1.72 A, is 0.833 A. The waveforms for the diode voltages D1, D2, D3, and D4 are shown in Figures 19. The highest voltage stress across the diodes is found to be 78 V. The output voltage waveform that occurs is seen in Figure 20. Typically, the output voltage is 138 V. The test findings are consistent with the analysis and modelling of the converter. Figure 21 displays the efficiency of the converter.

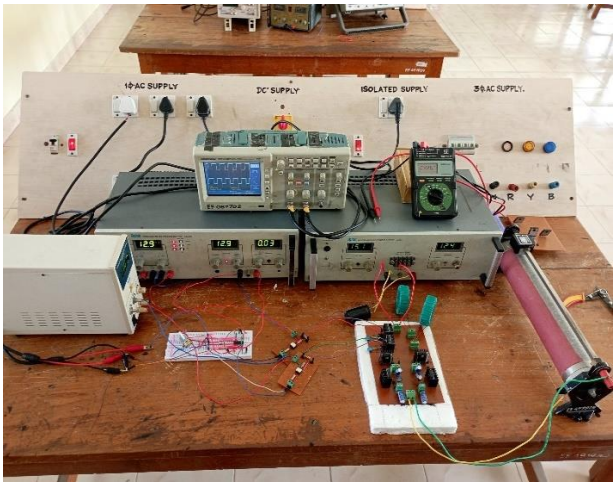

Fig 14: Hardware setup

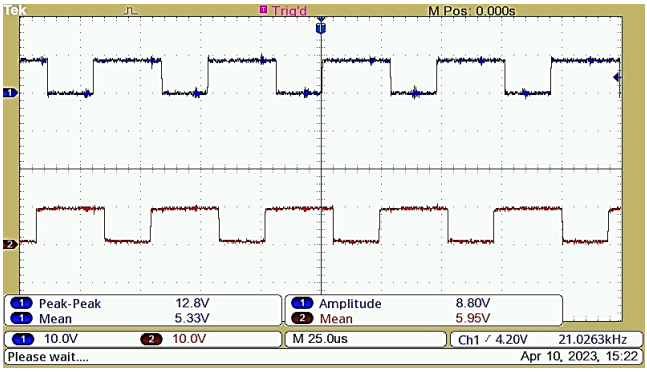

Fig 15: Gate Pulses

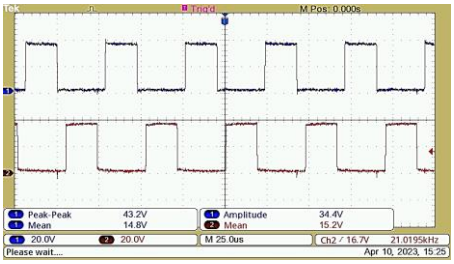

Fig 16: Voltage across the IGBT S_1 and S_2

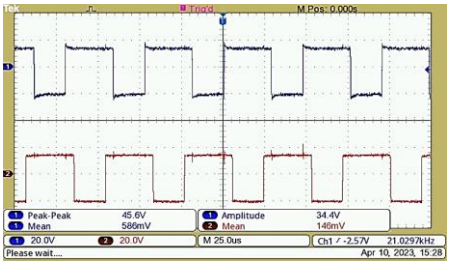

Fig 17: Voltage across the inductors L_1 and L_2

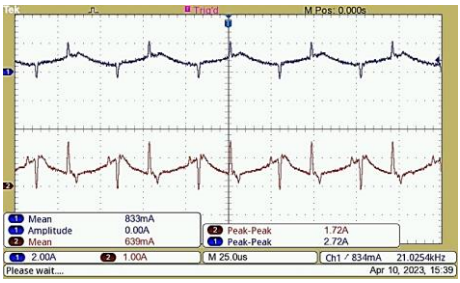

Fig 18: Current through the inductors L_1 and L_2

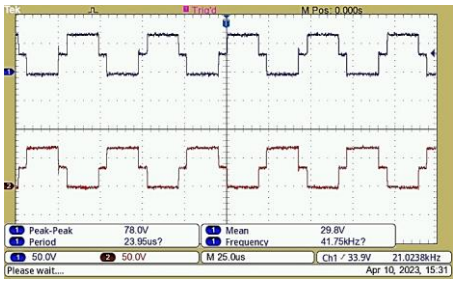

Fig 19: Voltage across the diodes D_1, D_4 and D_2, D_3

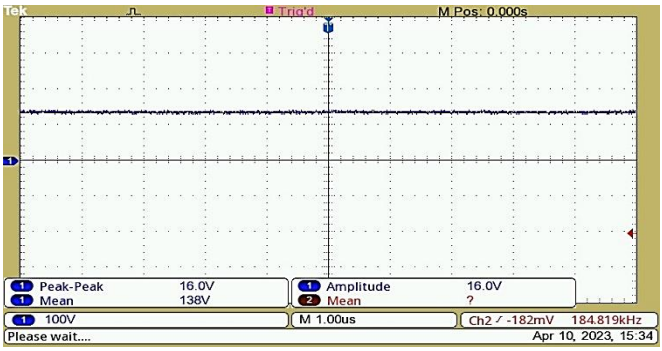

Fig 20: Output voltage

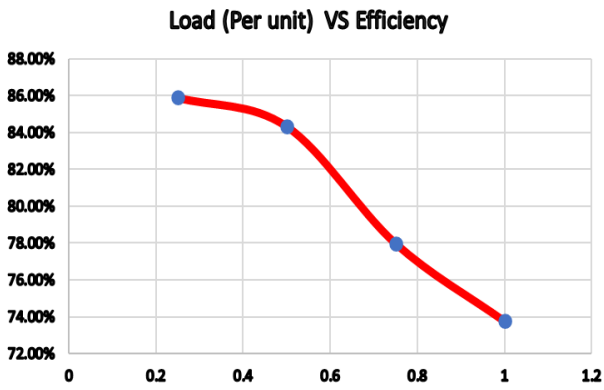

Fig 21: Efficiency of the converter

At 20% load, the converter efficiency is around 86% and when the load increases, correspondingly the converter efficiency decreases due to abrupt increase in source current. This problem occurs because of the inductor. There are some spikes in the inductor current when the duty cycle is greater than 60%. But this problem can be avoided when the duty cycle is 50%. However, the boost factor is the main constraint when the duty cycle is reduced. So, care must be taken while designing the inductor.

5 CONCLUSION

The voltage multiplier circuit and interleaving topology of a DC-DC converter boost converter are given. The calculation and explanation of the operation mode for static voltage gain. The components voltage stress and input current ripples are depicted in the simulation. The theory and method were tested with a hardware prototype. Due to its low ripples and continuous input current, this converter can be used with renewable energy applications. The voltage stress that the active switches and diodes suffer is quite modest when compared to a conventional boost converter. Despite all of its advantages, there are still some challenges with this converter. For instance, the feedback loop's design is complicated by the floating output voltage. Additionally, the voltage multiplier circuit's output can have a lot of ripples.

7 REFERENCES

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